

Biology — Exam 02 (Ch 7: Metabolism)

SINGLE ATTEMPT OVERVIEW

Re-sit Attempt | 2026-05-02 | Federal Board Class IX | Biology exam-02-ch7
Syllabus: Chapter 7 — Metabolism (Sections 7.1–7.8) | Previous sit: 2026-04-30 (93.1%)

PERFECT SCORE — 65 / 65 (100%) A+

2nd perfect score overall (after Math exam-01, 2026-04-18) | 8th consecutive perfect 12/12 MCQ score

65/65

TOTAL SCORE

100%

PERCENTAGE

12/12

MCQS (SEC A)

33/33

SHORT (SEC B)

20/20

LONG (SEC C)

A+

GRADE

Section Breakdown

Section A — MCQs	12/12 (100%)
Section B — Short Answers	33/33 (100%)
Section C — Long Answers	20/20 (100%)

Question-by-Question Summary

Q	Topic / OR choice	Marks	Result	Notes
2(i) OR	Why enzymes are biological catalysts; 2 characteristics	3	3/3	Both characteristics with evidence
2(ii)	Three types of cofactors with examples	3	3/3	All three named, described, exemplified
2(iii) OR	ES and EP complex — 3-step mechanism	3	3/3	All three steps with correct complex names
2(iv)	Effect of temperature on enzyme activity	3	3/3	All three zones; denaturation/reversibility correct
2(v) OR	Allosteric site + non-competitive inhibitor mechanism	3	3/3	Allosteric site explicitly named — weakness FIXED
2(vi) OR	ATP-ADP cycle: energy storage and release	3	3/3	Condensation/hydrolysis + significance
2(vii) OR	Three limiting factors of photosynthesis	3	3/3	Light, CO ₂ , temperature — all with mechanisms
2(viii)	Light-dependent reactions: definition, location, products	3	3/3	Thylakoid membrane; ATP/NADPH/O ₂ products
2(ix) OR	Aerobic respiration equation + ATP yield per glucose	3	3/3	Equation correct; 2+2+32=36 breakdown; ETC weakness FIXED
2(x) OR	Role of Krebs cycle; products formed	3	3/3	Mitochondrial matrix; 4CO ₂ +2ATP+6NADH+2FADH ₂
2(xi) OR	Alcoholic vs lactic acid fermentation comparison	3	3/3	Table format; organisms, products, equations, examples
3(i) OR	Three factors affecting enzyme activity (temp, pH, substrate)	5	5/5	Vmax named; pepsin/trypsin pH correct; inhibitors included
3(ii)	Mechanism + activation energy + lock-and-key vs induced-fit	5	5/5	Emil Fischer; rigid vs flexible; conformational change
3(iii)	ATP as energy currency: structure + cycle + uses	5	5/5	100–150 mol/day; AMP/ADP/ATP hierarchy; diagram drawn
3(iv) OR	Aerobic respiration: 4 stages with ATP yield	5	5/5	Inner mitochondrial membrane for ETC — weakness FIXED
TOTAL		65	65/65	Perfect score — no deductions

Comparison: Re-sit vs April 30 Attempt

Section	2026-04-30	2026-05-02 (Re-sit)	Change
Section A MCQs	12/12	12/12	—
Section B Short	29.5/33	33/33	+3.5
Section C Long	19/20	20/20	+1.0

Total	60.5/65 (93.1%)	65/65 (100%)	+4.5 marks (+6.9%)
All three weaknesses from the April 30 report corrected: allosteric site named (Q2v), ETC location stated (Q2ix/3iv), mechanism-behind-concepts pattern resolved.			

Biology Subject Performance Card

exam-01-ch1-6 (2026-04-03):	52/65 (80.0%)
exam-02-ch7 (2026-04-30):	60.5/65 (93.1%)
exam-02-ch7 re-sit (2026-05-02):	65/65 (100%)
Biology average (best per exam):	90.0% (52+65)/130

Key Strengths (This Attempt)

- 8 consecutive perfect MCQ scores
- All previous weaknesses self-corrected
- Vmax named for substrate concentration
- pH values for pepsin (2.0) and trypsin (8.0) from memory
- Induced-Fit: conformational change correctly explained
- ETC: NADH→3ATP, FADH₂→2ATP ratios given
- ATP-ADP cycle drawn with condensation/hydrolysis
- All 15 sub-questions answered, none omitted

Revision Targets for Ch 8 Onwards

- No current weaknesses from this paper
- Apply location-naming discipline to new topics (xylem/phloem/stomata)
- Always write overall balanced equations in long answers even when also stated in short
- Watch for new chapter patterns in plant physiology (osmosis, transpiration pull)